SUMMARY OF PREVIOUS & CURRENT RESEARCH

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1. Constraining the end of reionization with deep Lyα spectroscopy

Lyα Emission as a Probe of Reionization: Reionization is the last major phase transition of the intergalactic medium (IGM), and scrutinizing the detailed evolution of the IGM is a key frontier in observational cosmology, which also provides key information for studying the dominant source of the ionizing photons, galaxies in the early universe. Wilkinson Microwave Anisotropy Probe and Planck observations constrain the midpoint of reionization to be $z \sim 8-9$ (Larson et al. 2011, Planck Collaboration et al. 2016), and quasar observations at high redshift suggest that it ended at $z \sim 6$ (e.g., McGreer et al. 2015). Complementary measurements of the Lyα emitter (LAE) luminosity function at $z \sim 6$ agree with those from the CMB and quasar observations (e.g., Malhotra & Rhoads 2004; Konno et al. 2018). However, robust studies of the IGM during reionization are still limited, as it is, for example, difficult to map the neutral fraction of the IGM during reionization with quasar spectroscopy due to the lack of a large population of quasars at $z > 7$.

A presently accessible method for studying the IGM in the reionization era is to observe Lyα emission from continuum-selected galaxies with follow-up spectroscopy. The fraction of Lyman break galaxies with strong Lyα detection (known as the "Lyα fraction") was found to increase from $z = 3 \rightarrow 6$ (Stark et al. 2010). It was thus to be expected that the Lyα fraction at $z \sim 7$ would be at least as high as at $z = 6$. However, initial studies have found an apparent deficit of Lyα emission at $z > 6.5$ (e.g., Fontana et al. 2010; Pentericci et al. 2011, 2018), and the perceived drop in Lyα emission at $z > 6$ may imply that the neutral hydrogen (HI) fraction in the IGM increases significantly from $z \sim 6 \rightarrow 7$, although other galaxy evolutionary features and the uncertainty of the Lyman continuum escape fraction need to be taken into account (e.g., Dijkstra 2014; Kimm et al. 2017; Rosdahl et al. 2018).

Despite this tantalizing evidence, measuring the Lyα fraction depends on the sensitivity of the observed spectra and the completeness of the detected LAEs. For instance, selection criteria without longer wavelength continuum detection would lead to a bias toward faint galaxies with strong Lyα emission (see discussion in De Barros et al. 2017). Also, Caruana et al. (2018) showed that the choice of a rest-frame equivalent-width (EW) cut seems to be impactful on the measured Lyα fraction, finding no dependence of the Lyα fraction on redshift at $3 < z < 6$ with no EW threshold (commonly 25Å).

All things considered, while many previous studies have used the Lyα fraction as a measure of the evolution of Lyα emission, it is a somewhat less constraining measure, since it often does not fully account for the continuum luminosity of LAEs as well as it assumes that all non-detection objects among the observed galaxies are at that redshift, which is not necessarily to be true without spectroscopic confirmation. ***To provide the unbiased tracer of HI in the IGM using Lyα emission, I use the Lyα EW distribution during the epoch of reionization as a probe of the IGM HI fraction by applying the newly-introduced methodology in my recent publication (Jung et al. 2018), which accounts for data incompleteness (e.g., observing conditions, galaxy ultraviolt(UV) continuum luminosity, and galaxy photometric redshift***

probability), to a comprehensive spectroscopic dataset of Lyα emission.

Ground-based Spectroscopic Data for Lyα Emission during the Epoch of Reionization: To search for Lyα emission from galaxies in the reionization era, I have performed deep spectroscopic observations of candidate galaxies from the Cosmic Assembly Near-infrared Deep Extragalactic Legacy Survey (CANDELS). I use the galaxy sample from Finkelstein et al. (2015), which consists of ~ 1000 candidate $z > 6$ galaxies in the CANDELS GOODS-S and GOODS-N fields, as well as the Hubble Ultra Deep Field. A unique aspect of my program is that, compared to other studies which use only optical or NIR spectrographs, I have data from both the DEIMOS (optical) and MOSFIRE (NIR) spectrographs on the Keck 10-meter telescopes. This is *required* for strong constraints on the observability of Lyα emission from galaxies at $z_{photo} \sim 7$, as the broad range of the photometric redshift probability distributions of candidate galaxies results in a significant chance of the true redshift being anywhere in the range of $6 < z < 8$ (see the left panel in Figure 1). **Therefore, spectroscopic follow-up with both DEIMOS and MOSFIRE is necessary for ensuring comprehensive wavelength coverage of Lyα emission at $z \sim 6-8$, and is thus crucial to probing into the evolution of reionization.**

The entire observing program of this study is a total of 18 nights of spectroscopic integration for 183 galaxies at $z > 5$: for 118 galaxies with Keck/DEIMOS (PI: R. Livermore) and 84 galaxies with Keck/MOSFIRE (PI: S. Finkelstein) (the majority coming through the NASA/Keck allocation). Spectroscopic observation details are shown in Figure 1 (right).

Unbiased Measure of the Lyα Equivalent-width Distribution as a Probe of Reionization: Utilizing my deep spectroscopic dataset, I place constraints on the evolution of the Lyα EW distribution, using detailed simulations to include the effects of data incompleteness. This distribution function has been well studied at $0.3 < z < 6$, and has been found to have the form of an exponential distribution, dN/dEW $\propto$ exp(-EW/W_0), with a characteristic EW e-folding scale (W_0) of $\sim$ 60Å over the epoch $0.3 < z < 3$ (e.g., Gronwall et al. 2007; Guaita et al. 2010, Ciardullo et al. 2012, Wold et al. 2014, 2017); and possibly higher at higher redshift (Ouchi et al. 2008, Hu et al. 2010, Kashikawa et al. 2011, Zheng et al. 2014, Hashimoto et al. 2017). Particularly, in the epoch of reionization, the HI atoms in the IGM are expected to diminish these EWs, preferentially putting stronger impact on UV-faint sources and lowering the e-folding scale (W_0) of the observed Lyα EW distribution by rejecting high EW sources (e.g., Bolton et al. 2013, Kakiichi et al. 2016, Mason et al. 2018). This gives us our research question: at what confidence can new observations rule out the e-folding scale (W_0) of $\sim$ 60Å at $z > 6$?

My first study based on our DEIMOS observations (118 galaxies at $5 < z < 7$) has been published in Jung et al. (2018). I introduced my new scheme to constrain the Lyα EW distribution by considering the sources of data incompleteness, the photometric redshift probability and observational conditions (e.g., observational depth, wavelength coverage, and sky emission). I simulate the expected number of detections with the various shapes of Lyα EW distributions, and fit the constructed probability distribution of the expected number of detections to the actually observed detections to derive the probability distribution of e-folding scale (W_0) of the Lyα EW distribution. This brings the least biased measure of the Lyα EW distribution yet. **My first result of the measure of the e-folding scale (W_0) of the Lyα EW distribution at $z \sim 6.5$ rejects the Lyα EW distribution with $W_0 > 36.4$Å (125.3Å) at 1σ (2σ) significance (Figure 2) which provides the**

additional evidence that the Lyα EW distribution declines at $z > 6$, suggesting an increasing HI fraction in the IGM. Due to the small number of detections from our DEIMOS observation, our current measure is less constraining, but the additional detections (e.g., my in-hand MOSFIRE data) will be combined in my future study during the fellowship years, providing a robust measure of the Lyα EW distribution at the epoch of reionization.

2. Spatially-Resolved Stellar Population Study for $z \sim 4$ Galaxies

Before I started to work on Lyα emission as a probe of reionization, I performed spatially-resolved stellar population study using *Hubble Space Telescope (HST)* images. Understanding the true nature of star-formation regulation in the galaxies is the key to understand the drivers behind the evolution of the global star formation history through the cosmic time. Recent observational discoveries provide evidence that the most massive galaxies at $z \sim 2$ likely begin to quench star formation, forming bulges (e.g., Barro et al. 2014, Nelson et al. 2014). More detailed analysis of the evolution of massive galaxies at $z \lesssim 2$ has been done in a spatially-resolved manner, introducing a so-called *inside-out growth* scenario that massive galaxies grow outward, forming stars in their outskirts while the central components are less active at star formation (e.g., Nelson et al. 2016).

Investigating the resolved star formation activity at higher redshift universe, I published my first PhD project, *Evidence for Reduced Specific Star Formation Rates in the Centers of Massive Galaxies at $z = 4$* ApJ, 834, 81, Jung et al. (2017) which performed **the first-ever spatially resolved modeling of $z \sim 4$ galaxies**. I developed my own Markov Chain Monte Carlo stellar population modeling code, and provided the first evidence for inside-out growth at $z > 3.5$ that the galaxies with the highest central mass densities likely suppress star formation in their centers, compared to their outskirts. Then, what would star formation inside galaxies be like in even earlier universe? My research will be extended with *the James Webb Space Telescope (JWST)* at the end of my fellowship years.

3. Building Dark Matter Halo Merger Trees

During my master's program at Yonsei University, I studied theoretical understanding of galaxy evolution using N-body simulations and a semi-analytic model (SAM) with Prof. Sukyoung K. Yi. **Specifically, I developed a dark matter (DM) halo merger tree builder, called (ySAMtm; Jung et al. 2014), which constructs DM halo merger trees from cosmological N-body simulations and provides a DM halo backbone for the SAM of Prof. Yi's group (ySAM; Lee & Yi 2013)**. As a merger tree developer, I attended the Sussing Merger Trees Workshop (Srisawat et al. 2013), performing a large comparison project of DM halo merger tree builders in which 10 different tree-building algorithms were applied to the same set of halo catalogues and the results compared. Additionally, in collaboration with Dr. S. Peirani at the Institut d'Astrophysique de Paris, I worked on predicting the evolution of the distribution of baryons (gas and stars) in a Local Group-type universe, using hydrodynamical zoom simulations in the standard Λ cold dark matter cosmology. The study suggests that physical mechanisms able to drive the gas out of the virial radius at high redshifts will have a stronger impact on the deficit of baryons in the mass budget of Milky Way-type galaxies at present times, and this project was published in Peirani et al. (2012). In the extended study of my master's thesis at Yonsei, published in Jung et al. (2014), I examined whether large-scale environment affects the mass assembly history of central galaxies, analyzing model galaxies in a semi-analytic model.